

From Evaluation to Feedback: A Feature-Based and LLM-Constrained Tool for Korean Writing Assessment

Chanwoo Jang
Inha University
Incheon, South Korea
grrlkk@inha.edu

Seokho Ahn
Inha University
Incheon, South Korea
sokho0514@inha.edu

Sungmin Chang
Inha University
Incheon, South Korea
mrch17@inha.ac.kr

Ganghee Go
Inha University
Incheon, South Korea
khko99@inha.edu

Myungsun Shin
Inha University
Incheon, South Korea
rescript@inha.ac.kr

Do-Guk Kim*
Inha University
Incheon, South Korea
dgtkim@inha.ac.kr

Jinyong Yun
Inha University
Incheon, South Korea
12191632y@inha.edu

Ho-Hyun Kil
Kongju National University
Kongju, South Korea
kilhh@kongju.ac.kr

Young-Duk Seo*
Inha University
Incheon, South Korea
mysid88@inha.ac.kr

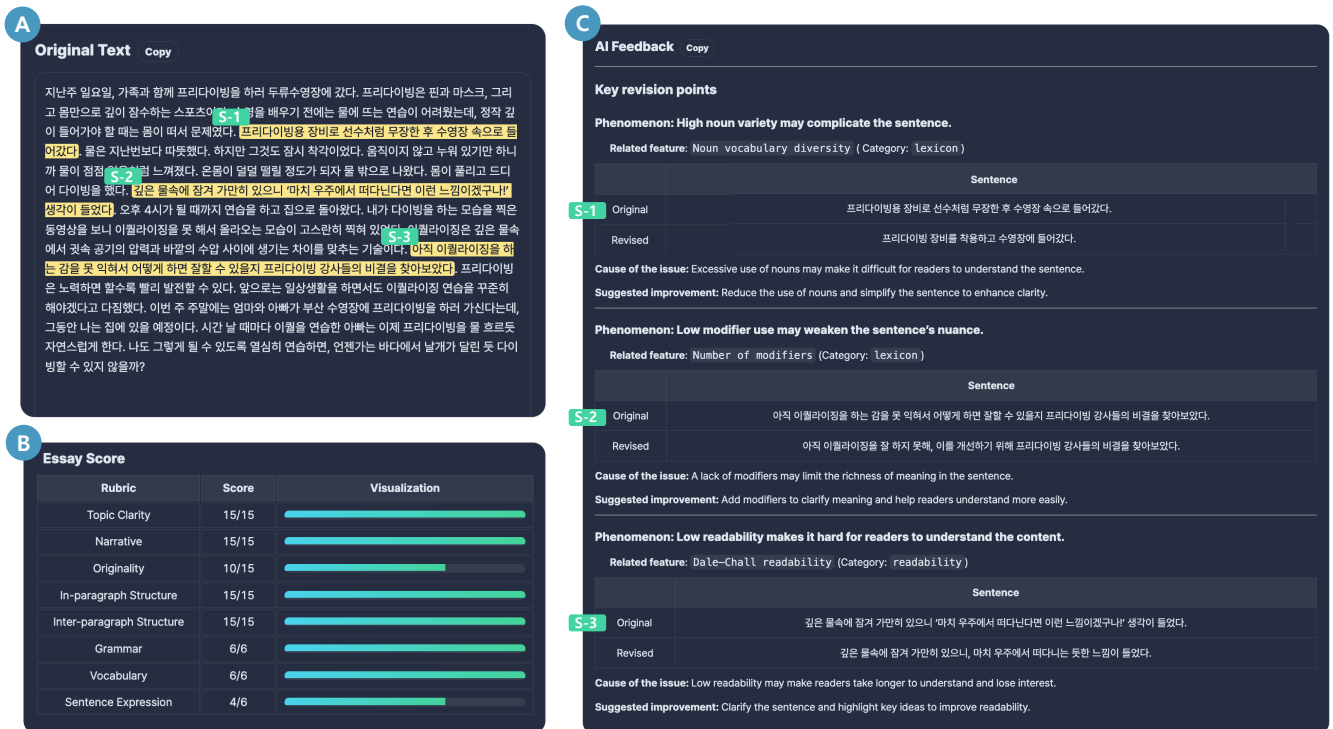


Figure 1: FEAk feedback example showing (A) input Korean essay, (B) computed rubric scores across eight dimensions, and (C) generated actionable feedback with evidence-based suggestions and revision examples.

*Co-corresponding authors.



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Abstract

Integrating LLMs with automated writing evaluation can transform scoring models into feedback tutors that translate diagnostic signals into actionable guidance. Ensuring reliable and consistent results in such systems requires addressing information overload, yet they still rely on manual intervention to extract key signals for meaningful feedback, limiting full automation. To address these challenges, we propose FEAk, a pipeline for Korean writing that dynamically identifies rubric-linked linguistic features with low values and uses them as evidence for LLM-based feedback generation. By grounding generative feedback in measurable diagnostic signals, FEAk generates interpretable and verifiable feedback. This design bridges quantitative evaluation with pedagogically meaningful guidance while operating autonomously. Experiments with automatic and human evaluations show that FEAk generates real-time feedback comparable to expert guidance and outperforms general LLM-based systems in quality and reliability.

CCS Concepts

• **Applied computing** → **Interactive learning environments**;
• **Computing methodologies** → *Natural language generation*;
Artificial intelligence.

Keywords

Automated Writing Evaluation, LLM-based Feedback Generation, Explainable AI in Education, Korean Writing Education

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1 Introduction

The goal of writing instruction is to help learners express their thoughts clearly and logically [11]. Formative feedback is widely recognized as the most effective means of achieving this goal [13]. However, in real world educational settings, providing consistent, timely, and specific feedback to all learners remains structurally difficult [9]. To address this challenge, extensive research has been conducted on Automated Writing Evaluation (AWE) systems [30]. Traditional AWE systems achieve high scoring accuracy by leveraging interpretable linguistic features such as lexical diversity and syntactic complexity [24]. Nevertheless, they largely function as *sophisticated graders* rather than tutors, as they fail to translate diagnostic results into actionable, sentence-level feedback that effectively supports revision. To bridge this diagnosis-to-action gap, recent studies have begun integrating large language models (LLMs) to generate feedback beyond simple grading.

Despite their generative capabilities, existing LLM-based feedback approaches face fundamental limitations. Methods that rely solely on rubric scores and original texts tend to produce overly general or inconsistent feedback, while incorporating all linguistic features at once leads to information overload and reduced reliability due to excessive input length. As a result, many systems

still depend on human intervention to identify and prioritize key feedback signals, undermining the goal of fully automated feedback. Furthermore, most prior work has focused on English, leaving a substantial gap for languages with complex morphology such as Korean. Directly applying English-centric models often fails to capture the linguistic and discourse-level nuances of Korean writing, highlighting the need for a language-specific approach.

To address these challenges, particularly in the context of Korean writing education, we propose **FEAk** (Feature-based Evaluation and AI feedback for Korean writing). FEAk dynamically identifies rubric-linked linguistic features with significantly low values and uses them as explicit evidence for LLM-based feedback generation. By grounding generative feedback in measurable diagnostic signals, FEAk ensures that each suggestion is interpretable, verifiable, and pedagogically meaningful, effectively integrating AWE's robust quantitative assessment with the generative strengths of LLMs.

FEAk operates in three stages. First, an enhanced AWE model analyzes the input essay to compute linguistic features and rubric scores (**Evaluation stage**, Section 3.1). Next, these features are compared against elite benchmark data to dynamically select rubric-relevant features that exhibit the largest deficiencies (**Feature Extraction stage**, Section 3.2). Finally, FEAk generates sentence-level feedback using a concise evidence sheet containing the selected feature values and their standardized gaps from benchmark distributions (**Feedback stage**, Section 3.3). Both automatic and human evaluations demonstrate that FEAk produces real-time feedback comparable to expert guidance while outperforming general LLM-based systems in quality and reliability.

Our contributions are summarized as follows:

- We propose FEAk, a framework for Korean writing that effectively integrates a refined AWE system with an LLM to bridge the gap between quantitative evaluation and generative feedback.
- FEAk dynamically selects priority linguistic features by comparing the current essay with elite benchmark data, enabling focused and adaptive feedback generation while reducing the need for external intervention.
- Experiments show that FEAk generates real-time feedback comparable to expert guidance, outperforming general LLM-based systems in both quality and reliability.

2 Related Work

In this section, we first review the evolution of AWE systems (Section 2.1). Next, we examine LLM-based feedback approaches, emphasizing emerging requirements: i) Explainable grounding, ii) Dynamic selection, and iii) Autonomous operation (Section 2.2). We finally review Korean feedback systems through these perspectives, highlighting their current challenges and gaps (Section 2.3).

2.1 Automated Writing Evaluation

Automated writing evaluation (AWE) has evolved through various scoring systems and methods. Early systems [18, 10, 3] quantified writing quality with interpretable features derived from linguistic indicators and coupled them with predictive models. Subsequently, deep learning models [30, 8, 33] and pre-trained language models [34, 32, 22] advanced scoring performance. Recent AWE research

has improved the accuracy and generalization of automated essay scoring through advances in neural and feature-based modeling.

UKTA [2] provides a unified Korean AWE framework that connects morpheme, feature, and rubric processing while reducing error propagation introduced by a Korean-specific morphological analyzer. [7] improved representation and generalization by combining grammatical diversity with error signals as auxiliary tasks in a multitask learning setup, while calibrating the scoring function with IRT. PLAES [5] employs a level-aware representation learning framework that enhances robustness to prompt variation and promotes cross-prompt generalization.

2.2 LLM-based Writing Feedback

The emergence of LLMs has shifted automated writing education from static evaluation to dynamic feedback [14]. Recent work demonstrates the educational effectiveness of LLM-based feedback and focuses on improving its quality and instructional value [6, 14].

In parallel, a line of research emphasizes reliability and evidence-based feedback [28]. In educational contexts, feedback is only meaningful when students can understand and trust it [15], a principle that aligns with Explainable AI (XAI), which aims to make AI decision-making processes transparent [1]. Motivated by this, SWIF²T [4] demonstrated the importance of explainable grounding by leveraging Retrieval Augmented Generation (RAG) to generate feedback based on objective facts from external scientific literature.

Another major research stream concerns the prioritization of feedback. [26] noted that while LLMs excel at local suggestions, they often overlook global issues like logical flaws in an entire text. Similarly, the *Lost in the Middle phenomenon* [23] highlighted the limitations of processing long contexts. Both studies underscore the need for feedback systems to dynamically identify and prioritize the most important features for improvement.

Finally, research aimed at securing the practicality and stability of feedback systems is also active. The issue of inconsistent LLM outputs poses a significant barrier to practical application [31]. PROF [25] introduced a dual LLM framework with a feedback generator and a student simulator, proposing a novel training loop that directly optimizes feedback performance through simulated learner revisions. This demonstrates that autonomous operation with minimal external intervention is a crucial system requirement. In sum, the core requirements for the current generation of LLM-based feedback systems converge on three principles: explainable grounding, dynamic selection, and autonomous operation.

2.3 LLM-based Korean Writing Feedback

While the above requirements—explainable grounding, dynamic selection, and autonomous operation—are well studied in English-dominant contexts, Korean writing education presents distinct challenges. Korean’s agglutinative morphology and complex discourse patterns complicate linguistic feature extraction, making accurate diagnosis more difficult than in English [2].

In Korea, the adoption of commercial LLM-based writing feedback services is rapidly increasing for use in educational settings [17]. KEEWiT¹ provides evaluation and feedback across six rubrics based on predefined foundational features such as essay length,

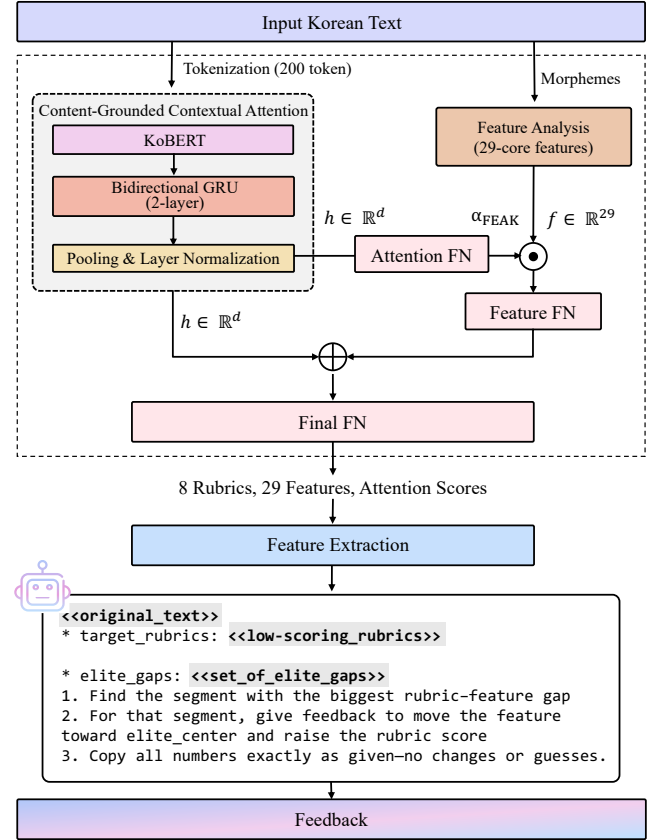


Figure 2: Illustrative overview of FEAk.

paragraph structure, and orthography. Jajakjajak² integrates rubric-based AI writing evaluation into the classroom workflow, while CLIPPO³ analyzes student writing based on grade level, subject-specific, or teacher-customized criteria to generate personalized feedback reports. While Korean commercial services offer practical advantages in classroom settings [19] and some improve transparency by showing scoring bases, they still lack the ability to dynamically select feedback items. Furthermore, because they largely rely on teacher-defined rubrics, they struggle to fully satisfy the requirements for objective, evidence-based feedback and autonomous operation without external intervention.

In contrast to the rapid growth of commercial services, Korean NLP research has focused mainly on evaluating the reliability of general-purpose LLMs such as ChatGPT within educational contexts [29, 16]. Other studies concentrate primarily on automated essay scoring rather than direct feedback generation [20, 21]. This limited progress can be largely attributed to the technical and linguistic challenges discussed earlier, which have substantially constrained the development of feedback-oriented systems in Korean [2]. Consequently, a systematic framework that satisfies these requirements remains absent in Korean feedback systems.

¹<https://keewi-t.korean.ai/>

²<https://jajakjajak.com/>

³<https://clipo.ai/>

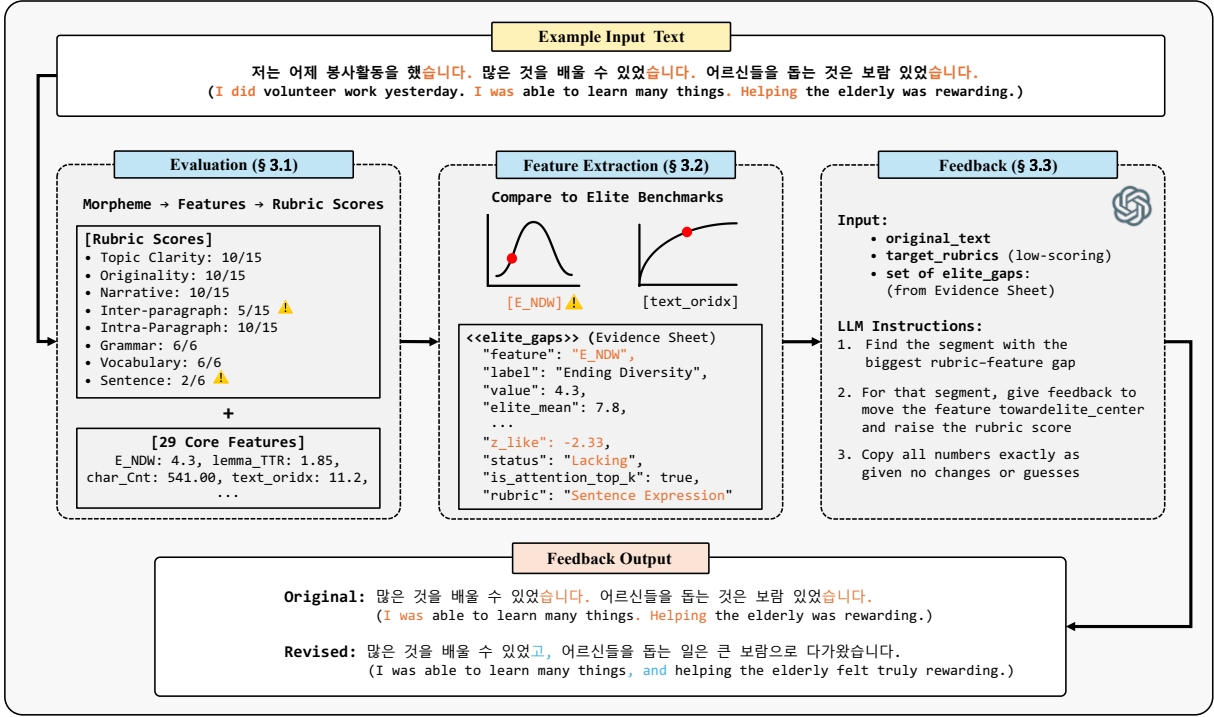


Figure 3: End-to-end example of the FEAK feedback generation process.

3 Methodology

This section describes the methodology of **FEAK** (Feature-based Evaluation and AI feedback for Korean writing). FEAK integrates quantitative diagnosis with evidence-grounded generation to achieve i) Explainable grounding, ii) Dynamic selection, and iii) Autonomous operation. Figure 3 demonstrates the complete pipeline through an example. The Evaluation stage (Section 3.1) computes linguistic features and rubric scores, establishing the foundation for explainable grounding. The Feature Extraction stage (Section 3.2) dynamically identifies priority features through elite gap analysis, fulfilling dynamic selection and enabling autonomous operation. The Feedback stage (Section 3.3) translates these objective diagnostics into student-comprehensible, sentence-level feedback.

3.1 Enhanced AWE Architecture

This section describes the enhanced AWE architecture, which quantifies linguistic features and rubric scores to provide interpretable diagnostic signals for feedback generation.

The Evaluation stage computes 29 linguistic features and 8 rubric scores from the input essay. This stage builds upon [2] but is re-trained on a subset of 29 features, retained from the original 294 to keep only the most relevant signals for the downstream dynamic prioritization. The selection criteria are detailed in Section 3.2.1. Key enhancements include expanded token capacity from 50 to 200, mean pooling over BiGRU hidden states, replacing the final-state representation, and Content-Grounded Contextual Attention that dynamically weights features based on essay semantics. The internal architecture is illustrated in Figure 2.

Given an essay with N sentences, pre-trained KoBERT⁴ encodes each sentence and a bidirectional GRU processes the sequence. The d -dimensional essay representation $\mathbf{h} \in \mathbb{R}^d$ is obtained by mean-pooling all hidden states followed by layer normalization. Simultaneously, 29 critical linguistic features $\mathbf{f} \in \mathbb{R}^{29}$ are computed via rule-based algorithms on morphologically analyzed text. These features include ending diversity, type-token ratios, and sentence overlap measures.

Evaluation stage then computes content-aware feature weights via contextual attention:

$$\alpha_{\text{FEAK}} = \text{softmax}(\mathbf{W}_{\text{attn}}\mathbf{h} + \mathbf{b}_{\text{attn}}) \quad (1)$$

where $\mathbf{W}_{\text{attn}}$ and $\mathbf{b}_{\text{attn}}$ are learnable attention parameters. Unlike prior models with fixed weights [2], these attention weights $\alpha_{\text{FEAK}} \in \mathbb{R}^{29}$ adapt to each essay's content, indicating which features are most salient for scoring this specific essay. Using these attention weights, the weighted features undergo linear projection and are concatenated with the essay representation:

$$\mathbf{h}_{\text{final}} = [\mathbf{h}; \mathbf{W}_{\text{proj}}(\alpha_{\text{FEAK}} \odot \mathbf{f})] \quad (2)$$

Finally, this integrated vector is passed to a fully-connected layer to produce rubric scores:

$$\mathbf{s} = \mathbf{W}_{\text{out}}\mathbf{h}_{\text{final}} + \mathbf{b}_{\text{out}} \quad (3)$$

where $\mathbf{s} \in \mathbb{R}^8$ contains scores for eight evaluation rubrics: Topic Clarity, Narrative Quality, Originality, Intra-paragraph Structure, Inter-paragraph Structure, Grammar, Vocabulary, and Sentence Expression.

⁴<https://github.com/SKTBrain/KoBERT>

The evaluation stage outputs three components used in subsequent stages: feature values $\mathbf{f} \in \mathbb{R}^{29}$, rubric scores $\mathbf{s} \in \mathbb{R}^8$, and Content-Grounded Contextual Attention weights $\alpha_{\text{FEAK}} \in \mathbb{R}^{29}$.

The high accuracy of this content-aware scoring model ensures the reliability of the subsequent diagnoses by the Feature Extraction and Feedback stages. The attention weights are reused by the Selector to prioritize features that are both statistically deficient via elite gap analysis and contextually salient for this essay.

3.2 Diagnostic-based Feature Extraction

This stage compares the linguistic feature scores of the current essay against elite benchmarks and dynamically identifies the core features most likely to yield the largest improvements in writing quality. By focusing on a small set of priority features, this stage filters out redundant information and concentrates the model's attention on the most interpretable signals for feedback generation, enhancing both clarity and reliability in the subsequent process.

3.2.1 Offline Core Feature Selection and Elite Benchmarking. Prior to deployment, we selected 29 core features from the full set of 294 by balancing diagnostic power with interpretability. This offline reduction removes highly correlated or redundant features and reduces computational cost, resulting in a cleaner and more coherent semantic space for reliable downstream selection.

The initial candidate pool was proposed in consultation with Korean linguistics experts and validated via five statistical procedures:

- (1) Univariate correlation significance (Pearson/Spearman)
- (2) Multivariate OLS significance (VIF filtering, HC3-robust)
- (3) Lasso stable selection (frequency ≥ 0.6)
- (4) Mutual information (top 25%)
- (5) Permutation importance (top 25%)

Features satisfying at least three criteria were reviewed by experts, yielding 29 final features. These selected features are grouped into six highly reliable dimensions for Korean writing assessment, as shown in Table 1.

To enable relative evaluation, FEAK compares each essay's feature scores against elite benchmarks derived from the top 10% of high-scoring essays. For each feature we computed and stored the elite mean μ_{elite} , standard deviation σ_{elite} , and decile values. The deciles serve as qualitative thresholds for interpreting feature states such as Lacking, Adequate, or Excessive. The set of 29 selected core features and their elite benchmarks are built offline prior to system deployment. The dynamic feature selection procedure for new essays is described in the next section.

3.2.2 Dynamic Feature Selection. Dynamic Feature Selector dynamically identifies six of the 29 features most likely to improve the essay through a three-step process: (1) detecting the lowest-scoring rubrics, (2) measuring each feature's deviation from elite benchmarks, and (3) ranking features by gap size, attention weight, and need for improvement to select the top six for feedback generation.

- **Step 1: Identify the two lowest-scoring rubrics.** The system selects the two lowest rubrics among the eight rubric scores produced by the Analyzer. For example, a low Sentence Expression score suggests issues with sentence-level fluency or variation.

Table 1: Core Features Categorized by Linguistic Unit

Category Level	Core Feature Examples
Basic Stats	Character count Word (eojeol) count
Morpheme/Word	Diversity of sentence-final endings Moving-average TTR of lemmas
Phrase/Clause/Sentence	Sentence function morpheme overlap Average sentence length in words
Paragraph	Paragraph noun overlap Average paragraph length in characters
Text	Average sentence similarity (SBERT [27]) Topic consistency (KeyBERT [12])
Applied Models	Dale-Chall readability index OR-index

- **Step 2: Quantify elite gaps for the 29 features.** For each core feature, the system computes a normalized deviation:

$$z_i = \frac{x_i - \mu_{\text{elite}}^{(i)}}{\sigma_{\text{elite}}^{(i)}} \quad (4)$$

where x_i is this essay's value for feature i . Large values of $|z_i|$ indicate substantial shortfalls relative to the elite group. For example, $z_i = -2.33$ suggests the essay is 2.33 standard deviations below the elite mean. This standardization allows features on different scales, such as character counts versus lexical diversity, to be compared on a common basis. Each feature's qualitative status is determined by deciles: for higher-is-better features, we label $x_i < p_{30}$ as Lacking, $p_{30} \leq x_i \leq p_{70}$ as Adequate, and $x_i > p_{70}$ as Excessive.

- **Step 3: Fuse multiple signals and set priorities.** The system aggregates multiple signals to assign a priority score to each feature:

$$\text{score}_i = |z_i| + \alpha_i + 0.2 \cdot \mathbb{I}_{\text{status} \neq \text{Adequate}}^{(i)} \quad (5)$$

The three components represent *gap magnitude*, *model salience*, and *improvement weight*. Gap magnitude $|z_i|$ captures the standardized deviation from the elite mean. Model salience α_i reflects the Evaluation model's attention weight for feature i , indicating contextual importance. An improvement weight of 0.2 is added when the feature's decile-based status indicates that revision is needed, emphasizing features that show room for improvement.

After scoring, we perform rubric-balanced selection targeting the two lowest scoring rubrics identified in Step 1. For each rubric, we use a pre-defined expert-mapped feature set from the offline setup. For example, Sentence Expression maps to features including ending diversity and average sentence length. We then select the top three features by priority score within each set. If fewer than six total are obtained, we take replacement from the global ranking.

This process yields up to six features, three per low-scoring rubric. Crucially, the Selector does not enumerate all features but dynamically chooses only those mapped to the essay's vulnerable rubrics. In this way, the system quantitatively determines which rubric, which feature, and by how much each should be improved.

- **Final output.** The system produces a structured evidence sheet containing the selected features and diagnostic metadata: current value, elite mean, z-score, qualitative status, target rubric, and expert guidelines. This evidence sheet then serves as the input to the feedback stage (see Figure 3).

These explicit justifications distinguish our approach from systems that report only rubric scores. A bare score like 2 out of 6 only signals weakness, whereas the evidence sheet provides concrete metrics (e.g., ending diversity notably below the elite average), enabling the feedback to produce verifiable guidance.

3.3 Evidence-Based Feedback Generation

The Feedback stage transforms the selected outputs into sentence-level feedback that students can understand and act upon. It receives six features, two weak rubrics, and the structured evidence sheet from the previous stage. The core design principle of the Feedback stage is to constrain the LLM to act as a translator rather than an evaluator. If the LLM were given the entire essay and asked to identify problems and generate feedback freely, it could hallucinate nonexistent issues or misjudge their importance. Instead, the stage uses the objective measurements computed in Section 3.1 and the selected features from Section 3.2 as explicit constraints, effectively narrowing the generation scope.

3.3.1 Constrained Prompting. The prompt combines three inputs: the original essay with numbered sentences, the two lowest rubrics identified in Step 1, and the evidence sheet for the selected six features. This structure is shown in the Generator input of Figure 3.

The prompt includes explicit instructions. First, locate the text segment showing the largest rubric-feature gap. Second, for that segment, generate feedback that moves the feature toward the elite mean and raises the rubric score. Third, all numerical values must be reproduced exactly as provided, without modification or inference.

The LLM leverages its surface-level pattern recognition capability. For example, if ending diversity is diagnosed as low, the Generator can detect repeated sentence-final endings without performing morphological analysis and cite those sentences as evidence. By separating evaluation from feedback generation, each feedback instance becomes auditable and traceable to quantitative evidence.

3.3.2 Three-Part Feedback Structure. Figure 1 illustrates a real example of FEAk's output interface, showing how the system converts an input essay into rubric-level evaluations and concrete sentence-level feedback. (A) displays the original essay with highlighted evidence sentences, (B) presents rubric scores across eight evaluation dimensions, and (C) shows the generated feedback containing evidence-linked suggestions and revision examples.

The generated feedback consists of three parts. The Phenomenon section explains what the problem is and why it matters, linking feature measurements to readability or rhetorical effectiveness. The Evidence section cites specific sentences by number that demonstrate the problem. The Revision section provides concrete before-and-after examples. Figure 3 presents an illustrative example of this structure, showing how the overall pipeline operates for a single feedback instance.

This structure separates diagnosis from prescription, ensuring that students first understand the problem before seeing the solution. Although the visible feedback appears concise and natural, each piece of advice is grounded in an extensive internal reasoning chain. Every suggestion can be traced through the Analyzer's feature computations, the Selector's elite-gap comparisons and z-scores, and the evidence constraints embedded in the Generator. Internally, all feature values and elite distributions are fully stored, allowing quantitative verification of every generated comment.

For example, while the student only sees user-friendly feedback such as "Vary your sentence endings for better flow," the system internally refers to quantitative indicators such as the ending-diversity score (4.3) and the corresponding z-score (2.33) relative to the elite benchmark. These values, along with corresponding entries in the evidence sheet, form the factual basis that supports each piece of feedback.

Overall, this pipeline bridges the gap between quantitative evaluation and meaningful feedback, integrating measurable linguistic evidence with controlled generation to produce feedback that is both interpretable and pedagogically meaningful.

4 Experiments

We evaluate FEAk through both user-based and model-based experiments to assess its effectiveness and reliability in real educational contexts. Specifically, we investigate the following research questions (RQs):

- **RQ1:** Does FEAk's evidence-based feedback improve writing quality compared to expert feedback and LLM feedback?
- **RQ2:** How do users perceive FEAk's feedback in terms of usefulness, specificity, trustworthiness, evidence alignment, and satisfaction?
- **RQ3:** Does the enhanced Analyzer achieve higher scoring performance than the baseline model?

4.1 Experimental Settings

To address the research questions, we conducted two experiments: a user feedback study (RQ1, RQ2) evaluating FEAk's feedback quality and educational impact, and a model evaluation (RQ3) assessing the model's accuracy and reliability.

4.1.1 User Feedback Experiment (RQ1, RQ2).

- **Participants.** We recruited 21 undergraduate students majoring in engineering and randomly assigned them to three conditions: Expert feedback, FEAk feedback, and general-purpose LLM feedback, with approximately seven participants per condition.
- **Task.** All participants were asked to write an argumentative essay on the topic "My good deed and what I learned from it", which was selected from the Korean high school writing curriculum. Using a single topic controlled for potential topic effects.
- **Procedure.** Each participant completed three stages in sequence: draft writing (15 minutes), feedback review (5 minutes), and revision (10 minutes). The total duration and sequence were identical across all conditions to ensure comparability.
- **Feedback Generation.** We controlled the feedback format, number of items (maximum three), and exposure duration across all

Table 2: Quality improvements by condition

Condition	Draft	Revision	Imprv. (Δ)	Time (s)
Expert	70.00 (5.76)	75.43 (5.43)	+5.43	471.20
LLM	68.14 (9.20)	70.29 (7.62)	+2.14	8.78
FEAK	67.43 (6.25)	71.86 (7.44)	+4.43	16.42
$F(2, 18)$	6.42** ($p = .008$)			

Table 3: User perception ratings by condition

Item	Expert	LLM	FEAK
Usefulness	4.07	3.43	3.83
Specificity	4.19	3.27	3.87
Trustworthiness	4.03	3.21	3.64
Evidence Alignment	3.94	3.09	3.78
Satisfaction	3.87	3.46	3.69
Mean	4.02	3.29	3.76

conditions. All feedback was generated with temperature set to 0.2, top- p to 0.9, and a maximum generation length of 512 tokens for consistency across conditions.

- **Expert feedback:** Produced by a Korean language education specialist and served as the human upper bound.
- **LLM feedback:** Generated by GPT-4o-mini using the same constrained prompt template as FEAK, serving as the baseline condition.
- **FEAK feedback:** Generated in the Feedback Generation stage using GPT-4o-mini as the generator LLM with a constrained prompt grounded in selected linguistic features and rubric-specific evidence, representing the proposed method.
- **Measures.** Three metrics were collected: (1) *quality improvement* ($\Delta = score_{rev} - score_{draft}$) computed using the AWE model described in Section 3.1 on a 0–100 scale, (2) *feedback generation time* (in seconds), and (3) *user perception scores* obtained from a five-item, five-point Likert-scale survey covering **Usefulness, Specificity, Trustworthiness, Evidence Alignment, and Satisfaction** [28].

4.1.2 Model Performance Evaluation (RQ3). To validate the enhanced Analyzer, we used the AI-HUB⁵ Essay Evaluation Dataset containing approximately 46,000 Korean essays from elementary to high school students, each scored by experts on 8 rubrics. We used 40,000 essays for training and 6,000 for testing. Evaluation metrics included Accuracy and Quadratic Weighted Kappa (QWK). The baseline for performance comparison was the UKTA model [2], which was reproduced under the same data and evaluation settings to ensure fairness.

4.2 Experimental Results

4.2.1 Feedback Quality and Efficiency (RQ1). Table 2 presents the quality improvements across feedback conditions. FEAK achieved an average improvement of +4.43 points, which was not significantly different from expert feedback (+5.43 points, $p = .231$) but

Table 4: Comparison of baseline and enhanced AWE models

Major-rubric	Rubric	Accuracy		QWK	
		UKTA	FEAK	UKTA	FEAK
Content	Topic Clarity	0.641	0.658	0.370	0.452
	Originality	0.621	0.647	0.172	0.318
	Narrative	0.622	0.644	0.488	0.553
Organization	Inter-paragraph	0.544	0.570	0.357	0.448
	Intra-paragraph	0.753	0.752	0.900	0.893
Expression	Grammar	0.601	0.597	0.280	0.279
	Vocabulary	0.643	0.649	0.343	0.360
	Sentence	0.715	0.724	0.883	0.868
Average		0.642	0.655	0.474	0.521

significantly higher than the general-purpose LLM (+2.14 points, $p = .028$). A one-way ANOVA revealed a significant main effect of feedback type on writing improvement ($F(2, 18) = 6.42$, $p = .008$, $\eta^2 = .42$). Tukey’s post-hoc test confirmed that both expert and FEAK feedback produced significantly greater gains than the LLM condition (Expert vs. LLM: $p = .006$; FEAK vs. LLM: $p = .028$), whereas the difference between Expert and FEAK was not statistically significant.

In terms of efficiency, FEAK generated feedback in an average of 16.4 seconds per essay, approximately 25 times faster than human experts (471 seconds) while maintaining comparable quality. Although the general-purpose LLM provided the fastest response (8.8 seconds), its improvement effect was substantially smaller. These results indicate that FEAK achieves near-expert feedback quality with dramatically higher efficiency, suggesting strong potential as a practical and scalable solution for real educational settings.

4.2.2 User Perception (RQ2). Table 3 presents user ratings across five perception dimensions. Expert feedback received the highest overall ratings, especially in specificity and trustworthiness, which may reflect its more detailed and context-sensitive nature. FEAK feedback was rated moderately high, outperforming the general-purpose LLM in all dimensions, especially in evidence alignment and usefulness, where participants appreciated the clarity of its explanations. The general-purpose LLM feedback received lower overall ratings, though its satisfaction score was relatively higher, likely due to its brevity and readability. Overall, these results indicate that FEAK provides concrete, evidence-based feedback that users perceive as informative, reliable, and pedagogically useful.

4.2.3 Analyzer Performance (RQ3). Table 4 compares the enhanced Analyzer against the baseline UKTA model. The enhanced version improved Accuracy on 6 of 8 rubrics and QWK on 6 of 8 rubrics. Substantial improvements were observed in the Content category (Topic Clarity, Originality, Narrative) and Organization category (Inter-paragraph Structure). Overall, Accuracy improved from 0.642 to 0.655 (+1.3pp) and QWK from 0.474 to 0.521 (+4.7pp). These improvements demonstrate that architectural enhancements including mean pooling, layer normalization, and content-aware attention positively impact rubric scoring accuracy. Reliable diagnostics are essential for downstream stages, as inaccurate scoring would propagate errors through the entire pipeline.

⁵<https://www.aihub.or.kr/>

5 Conclusion

We introduced FEAK, a pipeline that bridges the diagnosis-to-action gap by integrating interpretable feature-based diagnosis with constrained LLM generation. The system comprises three stages: an enhanced Analyzer computing linguistic features and rubric scores, a dynamic Selector prioritizing key features through statistical comparison, and a constrained Generator producing sentence-level feedback grounded in verifiable evidence. Experiments with 21 participants demonstrate that FEAK achieves quality improvements comparable to expert feedback while being faster and markedly outperforming general-purpose LLMs. Future research will address generalization across languages and genres, inclusion of higher-level discourse features, and examine long-term educational impact.

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